

Physics

1. Derivation of Lagrange's equations

1. Consider a system of particles: $k = 1, \dots, N_P$. Let $r^{(k)}$ be the position of the k^{th} particle relative to an inertial reference frame. Let $f^{(k)}$ be the resultant force acting on the mass m_k of the k^{th} particle.
2. Then, from Newton's Second Law: